

Philosophy of Space and Time

Lecture 9: *Introduction to Spacetime*

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Today's Plan

- 1 Recap and Setup
- 2 Geroch's Operational Turn
- 3 The Speed of Light
- 4 Binary Stars
- 5 The Conflict with Galilean Spacetime
- 6 Light Cones
- 7 What's New and What's Preserved
- 8 Summary and Preview

Where We Are

We have argued for **Galilean spacetime** as the right background for Newtonian physics:

- Simultaneity, temporal metric, spatial metric, affine structure
- No absolute velocity; acceleration survives as worldline curvature
- Spacetime symmetries (including boosts) match dynamical symmetries

?? *Does Galilean spacetime still postulate more structure than the physics needs?*

Two Threads

Thread 1 (Physics): There is a fact about the speed of light—it is the *same* for all inertial observers. This cannot be accommodated within Galilean spacetime.

Thread 2 (Methodology): Geroch's operational approach: build spacetime from the simplest primitives, minimize theoretical assumptions. Most basic: *light signal*.

These threads converge: once we take light signals as the primary tool for exploring spacetime, we can see how that leads to a different account of spacetime structure.

Geroch's Operational Turn

Building Spacetime Operationally

Geroch's strategy (Ch. 5): rather than *postulating* spacetime structure, *construct* it from what observers can do.

- Start with **events**—what happens, where and when
- An **observer** traces a worldline through events
- To describe relations between events, we need some physical process that connects them

?? *What is the simplest, most reliable physical process we can use to explore spacetime?*

Two Candidates: Rods vs. Light Signals

Measuring Rods

- Concrete, familiar, intuitive
- But: a rod is a *complicated* physical system
- Subject to thermal expansion, stresses, vibrations

Light Signals

- Propagate in vacuum, no internal structure
- Behavior described by a simple, universal law
- The *simplest* signal nature provides

Geroch's Choice

Take **light signals** as the fundamental tool. Build up the structure of spacetime from what observers can establish using clocks and light.

Why This Matters Philosophically

- Choosing the simplest primitive minimizes theoretical commitments; light signals minimal
 - Bonus: the same toolkit will work in curved spacetime (General Relativity), where rigid rods become problematic
- ?? *If primitive tools behave differently from what our classical intuitions suggest, our picture of spacetime will have to change.*

The Basic Toolkit

Geroch equips each observer with:

1. A **clock**—a device that measures elapsed time along the observer's worldline
2. The ability to **send and receive light signals**

From these two capacities alone, we will be able to:

- Determine distances to other events (by radar: send a signal, time the return)
- Synchronize clocks at different locations
- Extract the full geometric structure of spacetime

The Speed of Light

A Finite, Measurable Speed

Before pursuing the operational program, one empirical input: *what is the behavior of light?*

- Light has a finite speed (**Rømer** 1676: from delays in the eclipses of Jupiter's moons)
- Modern value: $c \approx 3 \times 10^8$ m/s
- This is *very fast* compared to other ordinary velocities
- But it is still *finite* rather than instantaneous

?? *Once we accept light has a definite, finite speed, we can ask a further question: does that speed depend on the motion of the source?*

Two Possibilities

Consider a lamp that emits light with speed c *relative to the lamp*. What speed do I measure, as an observer, if the lamp is moving toward me at speed v ?

Emission Theory

- Light speed *adds* to source velocity, like a bullet from a gun
- Measured speed: $c + v$
- Matches Galilean velocity addition
- Fits our classical intuitions

Source-Independent c

- Light speed is *fixed*, independent of source
- Measured speed: c , regardless of v
- Violates Galilean velocity addition
- Demands a revision of space and time

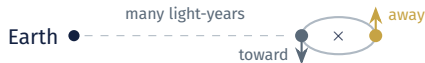
The choice between these is an *empirical* matter. And nature has a remarkably clear answer.

Binary Stars

The Setup: Eclipsing Binary Stars

Many stars are not solitary: they come in pairs, orbiting a common center of mass.

- In a typical orbit: one star is moving *toward* Earth while the other moves *away*
- Half an orbit later: the roles reverse
- These orbits are far enough away that the light takes *years* to reach us
- Orbital periods: days to years; distances: tens to thousands of light-years



The Prediction of Emission Theory

If light's speed depends on the source velocity, then:

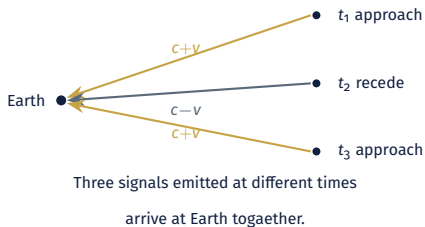
- Light from the star moving *toward* us travels at $c + v_{\text{orbit}}$
- Light from the star moving *away* from us travels at $c - v_{\text{orbit}}$
- Over a long journey, the faster light *catches up* with the slower light

Consequence: light emitted at *different* times in the orbit could reach us *simultaneously*.

What We Would See

If emission theory were true, light from different phases of the orbit would arrive jumbled together.

The observed binary would not trace out a simple ellipse.



What We Actually See

The Observation

Binary stars trace out *clean, periodic orbits*, perfectly described by Kepler's laws. No ghost images. No jumbled phases. No distortion.

- This was pointed out by **Willem de Sitter** in 1913
- He examined the light curves of known spectroscopic binaries
- If emission theory were correct, even a very small dependence on source velocity would produce enormous effects over the light-travel distance
- The effects are absent; the orbits look normal

?? *Conclusion: the speed of light does not depend on the velocity of the emitter, as established by routine astronomical observation.*

Why This Is a Powerful Argument

- The *enormous distances* act as an amplifier: tiny differences in light speed would produce visible effects after a journey of many light-years
- The effect would scale with distance: more distant binaries would be *more* distorted
- No observed binary shows this distortion—at any distance

Geroch's Methodological Point

Start with light signals as primitive; ask what they do; let observation tell us. The answer is unambiguous: light has a speed that does not depend on its source.

The Conflict

A Problem for Galilean Relativity

We now have two claims that *appear* to contradict each other:

1. **Galilean relativity**: the laws of physics are the same in all inertial frames, and velocities transform by simple addition
 2. **Constancy of c** : the speed of light is the same for all inertial observers, independent of source
- ??** *If velocities add in the Galilean way, then an observer moving toward a light source must measure a different speed than an observer at rest relative to the source. But observation says no.*

Three Ways Out

1. **Deny the principle of relativity.** Claim there is a preferred frame (the “aether”) in which c takes its standard value. Observers at rest in the aether measure c ; others measure different values.
2. **Deny the constancy of c .** Adopt an emission theory in which c depends on the source. (Ruled out by binary stars.)
3. **Revise our conceptions of space and time.** Give up Galilean velocity addition—which means giving up absolute simultaneity.

Option 3 is Einstein's. It requires a fundamental shift in how we think about simultaneity and duration.

A Note on aether Theory

Why not just accept option 1 (a preferred frame)?

- The 19th-century consensus: light is a wave; waves need a medium; that medium is the **luminiferous aether**
- Experimental test: set up an interferometer and look for changes as the Earth moves through the aether
- **Michelson & Morley** (1887): precise measurements produced a *null* result
- Every subsequent experiment has confirmed: no motion with respect to any aether can be detected

?? *We could posit an aether in principle, but we would have to postulate that all experiments designed to detect it are systematically fooled. Lorentz tried—but Einstein's move is cleaner.*

Break

Light Cones

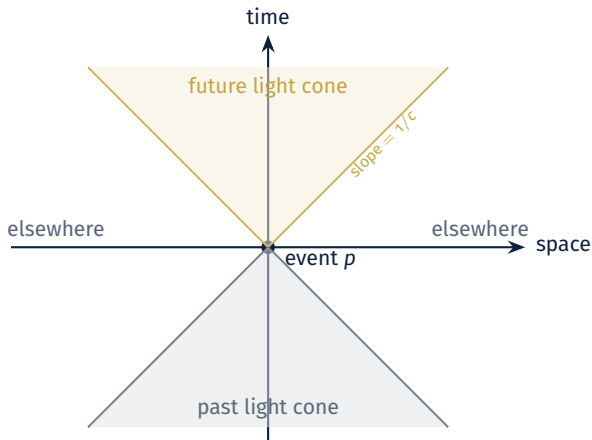
The Structure Light Signals Define

If we take light signals as our fundamental tool, what *structure* do they impose on spacetime?

- At each event, there is a definite collection of other events reachable by a light signal
- From event p : future light signals trace out a cone of “events we can reach at speed c ”
- Past light signals trace out the cone of “events that can send a signal to p ”
- Together these form the **light cone** at p

Because the speed of light is source-independent, this cone is an *objective* feature of the spacetime at each event—not observer-dependent.

The Light Cone at an Event



The light cone at p separates events into three regions: causal future, causal past, and “elsewhere” (neither reachable from p nor able to reach p).

Light Cones Replace Simultaneity

In Galilean spacetime, the fundamental relation is **simultaneity**: which events happen at the same time.

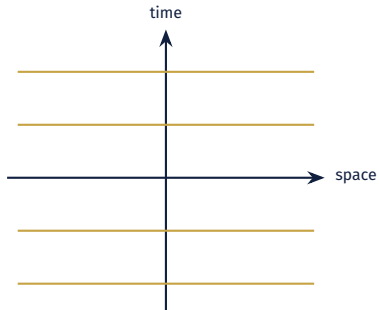
- Simultaneity slices partition spacetime into “layers of now”
- Every event has a unique “now”
- This is absolute and observer-independent

Light Cones Replace Simultaneity, 2

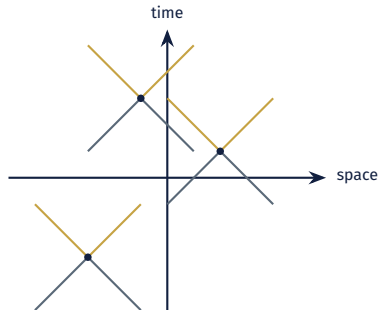
In the new picture, the fundamental relation is the **light-cone structure**:

- Each event has a future and past light cone
- This cone structure is absolute and observer-independent
- But there is no preferred way to slice spacetime into layers of “simultaneous events”

Two Pictures of Spacetime



Galilean:
simultaneity slices



Minkowski:
light cones at each event

The Trade

What Is Given Up

Absolute simultaneity: the claim that, for any two events, there is an objective fact about whether they occurred “at the same time.”

What Is Gained

A spacetime structure in which the speed of light is the same for all inertial observers—and in which the structure of space and time is unified into a single geometric object.

?? *Following the same pattern as before: we strip away one more piece of unobservable structure (absolute simultaneity), and the remaining structure (the light-cone structure) turns out to be exactly what the physics requires.*

What's New and What's Preserved

Preserved Features

Despite the dramatic change, much of our earlier picture survives:

- **Events** are still the primitives
- **Worldlines** still trace the histories of bodies
- **Inertial observers** are still those whose worldlines are straight
- **Affine structure** is preserved: straight vs. curved worldlines remain objective
- **Principle of relativity** holds: no inertial observer is privileged

The move from Galilean to Minkowski spacetime is, structurally, analogous to the move from full Newtonian to Galilean: strip away one more piece of excess structure.

New Features

What is *genuinely new* in the Minkowski picture:

- Simultaneity becomes **frame-dependent**: different inertial observers disagree about which events are simultaneous
 - Time intervals and spatial distances become frame-dependent
 - But a new invariant emerges: the **spacetime interval** (Geroch's main topic for Ch. 6; we'll develop this next week)
 - Space and time are no longer independent—they are aspects of a unified *spacetime*
- ??** *Hermann Minkowski (1908): "Space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality."*

Summary

1. **Geroch's operational turn:** build spacetime structure from what observers can do with clocks and light signals. Light is the simplest primitive—no internal structure, no material substratum required.
2. **The binary star argument:** observations of eclipsing binaries show that the speed of light does not depend on the velocity of the source. The universe runs the experiment for us.
3. **The conflict:** source-independent c cannot be reconciled with Galilean velocity addition. Something has to give.
4. **Einstein's move:** give up absolute simultaneity; replace simultaneity slices with light cones as the fundamental structure.
5. **The structural pattern:** each revision of spacetime drops structure the dynamics doesn't need. Full Newtonian $\rightarrow$ Galilean $\rightarrow$ Minkowski.

For Thursday

- **Read:** Maudlin, Ch. 4.
- **Think about:**
 - › If simultaneity is not absolute, what does it *mean* for two events to be “simultaneous in a given frame”?
 - › How can different observers disagree about simultaneity without contradicting each other?

Thursday: **Special Relativity and the Relativity of Simultaneity**

- Einstein's two postulates and their consequences
- A careful treatment of simultaneity using light signals
- How observers in relative motion disagree about which distant events happen “at the same time”
- Looking ahead to the exam (Tuesday, May 5)

Overflow: Why Rigid Rods Are Problematic

A subtle but important point:

- A “rigid body” is idealized as something whose shape is fixed by its internal constitution
- But any real rod is held together by forces between its parts
- These forces propagate at a finite speed (bounded by c)
- A truly rigid rod would transmit influences instantaneously—ruled out in relativistic physics
- So “rigidity” as a primitive is incompatible with relativity from the start

Using light signals instead sidesteps this difficulty. Light doesn’t need to be “rigid”—it just needs to propagate lawfully.

Overflow: Rømer and the Finite Speed of Light

Ole Rømer (1676): measured the speed of light from Jupiter's moons.

- Io's orbital period (around Jupiter) is very regular: about 42.5 hours
- But observed from Earth, the apparent period *varies*
- When Earth is moving toward Jupiter, eclipses appear earlier
- When Earth is moving away, eclipses appear later
- The variation reveals a finite speed: light from Io takes longer to reach us when we are farther away
- Rømer's estimate: close to the modern value of c

Before this, many (including Descartes) believed light propagated instantaneously.

Overflow: Michelson–Morley in More Detail

- Principle: split a light beam, send the two halves on perpendicular paths, recombine
- If the apparatus moves through the aether, the two paths take different times
- Observable: interference fringe shifts as apparatus rotates
- Expected shift (1887): well within experimental precision
- Observed shift: zero, to within experimental error
- Repeated many times with increasing precision—always null

Lorentz's response: postulate that motion through the aether causes physical contraction of the apparatus by just the right amount to cancel the effect. Einstein's response: drop the aether and rethink space and time.

Overflow: Why “Elsewhere” Matters

The light cone divides spacetime into three regions. The “elsewhere” region is a new feature compared to Galilean spacetime:

- Events in p ’s “elsewhere” cannot be causally connected to p
- No signal from p can reach them; no signal from them can reach p
- In Galilean spacetime, there are no such events: any event at a later time can in principle be influenced by any earlier event
- In Minkowski spacetime, the finite speed of light imposes a strict causal ordering

This has deep consequences—for causation, for the direction of time, and for cosmology. More on this in the coming weeks.

Overflow: The Binary Star Argument—Quantitative

How strong is the binary-star constraint?

- A typical orbital velocity: $v \sim 100 \text{ km/s}$
- If c depended on source: light speeds would differ by $\Delta c \sim v$
- For a binary at distance d , the time difference between fast and slow light arrivals: $\Delta t \sim d \cdot v/c^2$
- For $d = 100$ light-years: $\Delta t \sim$ days to months
- Orbital period: also days to months
- The effects would be *comparable to the period*—impossible to miss

De Sitter's original paper (1913): any emission theory predicting even a weak dependence of c on source velocity is ruled out by routine binary star data.

Questions?

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